SYSTEM HARDENING AND FIREWALL CONFIGURATION REPORT

Internship: Cognetix Technology
Task: Task 2 — Firewall Configuration & System Hardening
Platform: Kali Linux
Assessment Type: Local System Security Assessment
Student: Odai Mohsen Wahib

---

1. Introduction

The purpose of this task was to improve the security of a Kali Linux system by reviewing firewall settings, identifying listening network services, reviewing running background services, and checking the availability of system updates.

The assessment focused on reducing unnecessary network exposure while preserving the services required for normal system operation and cybersecurity lab activities.

2. Objectives

- Configure and review the UFW firewall.
- Restrict unnecessary incoming network connections.
- Identify listening ports and associated services.
- Review running services for possible hardening.
- Check available operating system updates.
- Test firewall behavior using Nmap.
- Document findings and recommend security improvements.

3. Scope and Tools

Scope: The Kali Linux virtual machine and its local firewall configuration.

Tools used:

- UFW — Uncomplicated Firewall
- Nmap — Network scanning and firewall testing
- ss — Listening port and socket inspection
- systemctl — Running service review

- APT — Package update management
- Docker CLI — Container status inspection

Only the local system and authorized lab network were considered in scope.

4. Methodology and Findings

4.1 Firewall Configuration

The UFW firewall was active during the assessment. The firewall status output showed an explicit inbound denial for FTP port 21 for both IPv4 and IPv6.

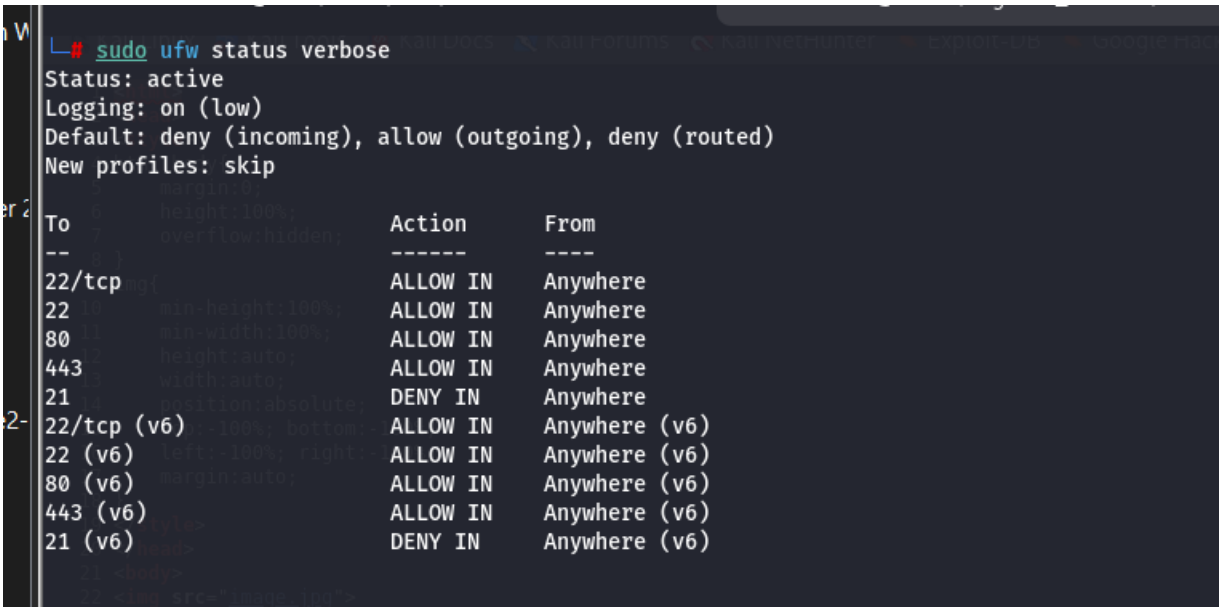
The earlier verbose status output also showed the following rules:

- SSH (22/tcp): Allowed
- HTTP (80/tcp): Allowed
- HTTPS (443/tcp): Allowed
- FTP (21/tcp): Denied

The later numbered-rule output displayed only the FTP denial rules. Therefore, the final rule list should be captured again before submission to confirm the current configuration.

The intended firewall policy is to deny incoming connections by default, allow outgoing connections, keep SSH available as requested, and block unnecessary services.

Screenshot 1 — UFW firewall status



4.2 Listening Ports and Network Services

The following listening ports were identified using ss -tulpen:

Port	Service	Observation
TCP 22	SSH	Listening on all IPv4 and IPv6 interfaces
TCP 5432	PostgreSQL	Bound to localhost only
TCP 1716	KDE Connect	Listening on all interfaces
UDP 1716	KDE Connect	Listening on all interfaces

SSH was retained because remote access on port 22 was intentionally required.

PostgreSQL was bound to the loopback addresses 127.0.0.1 and ::1, limiting its listening scope to the local system.

KDE Connect was running, but its intended use could not be confirmed. Its removal or disabling was therefore not performed as part of this assessment.

Screenshot 2 — Listening ports



4.3 Running Services Review

The running service list was reviewed using:

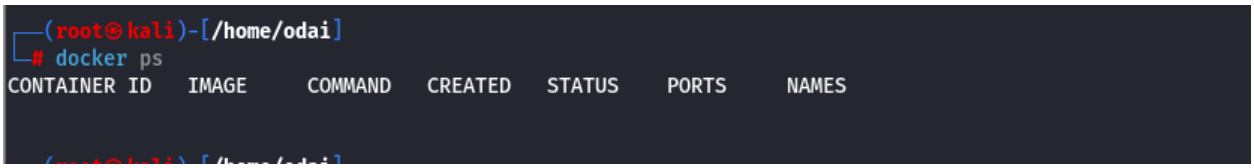
- systemctl --type=service --state=running

Twenty-eight loaded service units were listed. Core services such as NetworkManager, D-Bus, the GNOME Display Manager, OpenSSH, systemd components, and VMware tools were retained because they support networking, desktop operation, system management, remote access, or the virtual machine environment.

Docker and containerd were running, but docker ps showed no running containers at the time of inspection. PostgreSQL was also running. These services were not disabled because they may be needed for cybersecurity labs and development tools.

The assessment emphasized reviewing services before disabling them to avoid disrupting the desktop, networking, or lab environment.

Screenshot 3 — Running services and Docker status



4.4 System Updates

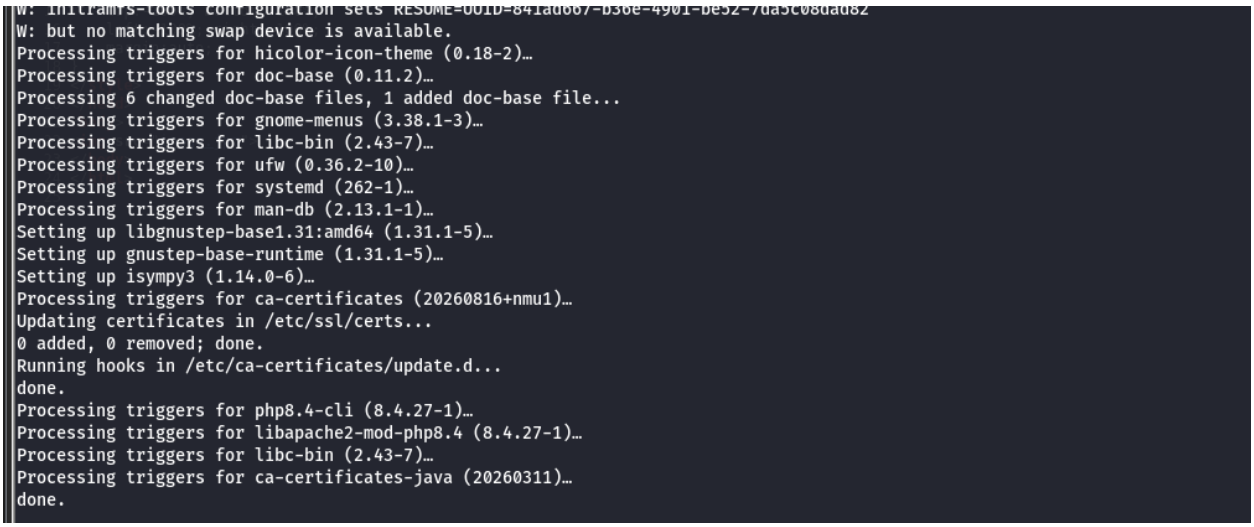
APT presented the following proposed package changes:

Item	Reported value
Packages to upgrade	203
New packages to install	15
Packages to remove	0
Packages not upgraded	608
Download size	475 MB
Additional disk space required	39.3 MB

The available disk space was reported as approximately 24.0 GB at the time of the upgrade prompt. No packages were proposed for removal in that summary.

The package summary was reviewed as part of system maintenance. A final completion message should be captured if the upgrade completed successfully.

Screenshot 4 — APT upgrade summary or completion



4.5 Firewall Testing with Nmap

The following command was used to test four TCP ports on the target host:

nmap -Pn -p 21,22,80,443 192.168.1.5

The reported results were:

Port	Service	Result
21/tcp	FTP	Filtered
22/tcp	SSH	Open
80/tcp	HTTP	Filtered
443/tcp	HTTPS	Filtered

The scan showed that SSH was reachable, while the other tested ports were filtered. A filtered result means Nmap could not determine whether the port was open because probes were blocked or did not receive a conclusive response.

These results do not, by themselves, prove that UFW was the only cause of filtering. The target IP and firewall configuration should be confirmed in the final evidence.

Screenshot 5 — Nmap firewall test

```
└─# nmap -Pn -p 21,22,80,443 192.168.1.5
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-28 19:15 +0530
Nmap scan report for 192.168.1.5
Host is up (0.00045s latency).

PORT    STATE    SERVICE
21/tcp  filtered ftp
22/tcp  open     ssh
80/tcp  filtered http
443/tcp filtered https

Nmap done: 1 IP address (1 host up) scanned in 1.81 seconds
```

4.6 Reboot Status

The system check reported No reboot flag found. This means the checked reboot indicator was absent; it does not guarantee that no update could ever require a reboot.

5. Risk Assessment

Finding	Risk level	Recommendation
SSH listening on all interfaces	Moderate, depending on network exposure	Keep SSH enabled as required; use strong authentication and restrict source access where practical
KDE Connect listening on TCP/UDP 1716	Low to moderate, depending on use and network	Disable or restrict it only if unnecessary
PostgreSQL running locally	Low network exposure based on observed binding	Keep bound to localhost unless remote database access is required
FTP port explicitly denied	Positive security control	Retain the deny rule if FTP access is not needed
608 packages listed as not upgraded	Requires review	Investigate held-back packages separately after the main upgrade
Docker service running with no active containers	Depends on intended use	Stop or disable Docker only if it is not required for lab work

Risk levels are preliminary and based on the limited local checks performed, not on a complete penetration test.

6. Recommendations

1. Confirm that UFW is active and that its default incoming policy is deny.
2. Keep SSH port 22 allowed as required, protect it with a strong unique password, and consider key-based authentication.
3. Keep unused ports and services blocked or disabled where appropriate.
4. Review KDE Connect and Docker based on actual usage before disabling them.
5. Complete and verify package updates; review packages that remain unupgraded.
6. Repeat the Nmap test from an authorized second machine and save the final output.

7. Avoid exposing PostgreSQL to external interfaces unless necessary.
8. Retain screenshots and command output as evidence for the internship submission.

7. Conclusion

The assessment reviewed the Kali Linux firewall, listening ports, running services, package update summary, and network filtering behavior. UFW was active, FTP port 21 had explicit inbound deny rules, SSH port 22 was reachable during the scan, and PostgreSQL was listening only on localhost.

The results identify several positive controls and areas for continued review. Final submission evidence should include the current verbose firewall status, a confirmed update result, and the Nmap test. Any hardening action that has not been verified should not be reported as completed.

End of Report